

Use Case: Make Restaurant Reservation

Primary Actor: Diner

Main Success Scenario:

1. Diner selects the "Make Reservation" option from the restaurant application. System displays the available dining dates, time slots, and live 2D floor plan.
2. Diner selects a desired date and dining time.
3. System displays the tables available for the selected dining window.
4. Diner selects an available table from the floor plan.
5. System validates that the selected table is still available for the chosen date and time.
6. System displays the restaurant menu and provides the option to pre-order food.
7. Diner selects menu items and quantities for the pre-order.
8. System calculates the estimated kitchen preparation time based on the selected food items.
9. System displays the reservation details, selected table, pre-ordered items, and estimated preparation time.
10. Diner confirms the reservation and pre-order.
11. System creates the reservation, marks the selected table as Reserved, and queues the pre-order for kitchen preparation.
12. System displays a reservation confirmation containing the table, date, time, pre-ordered items, and preparation timeline.
13. Use case ends successfully.

Alternate Flow

5a. Selected Table Becomes Unavailable

- 5a1. System detects that the selected table has been reserved by another diner for the same or overlapping dining window.
- 5a2. System displays a "Table No Longer Available" message.
- 5a3. System displays the updated floor plan with currently available tables.
- 5a4. Diner selects another available table.
- 5a5. System validates the newly selected table and continues from Step 6.
- 5a6. If no suitable table is available, the diner is prompted to select another time slot or date, and the reservation process is cancelled if the diner does not select one.